

# Cognitive Changes and Quality of Life in Neurocysticercosis: A Longitudinal Study

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**Background** Few studies have focused on the cognitive morbidity of neurocysticercosis (NCC), one of the most common parasitic infections of the central nervous system. We longitudinally assessed the cognitive status and quality of life (QoL) of patients with incident symptomatic NCC cases and matched controls.

**Methodology/Principal Findings** The setting of the study was the Sabogal Hospital and Cysticercosis Unit, Department of Transmissible Diseases, National Institute of Neurological Sciences, Lima, Peru. The design was a longitudinal study of new onset NCC cases and controls. Participants included a total of 14 patients with recently diagnosed NCC along with 14 healthy neighborhood controls and 7 recently diagnosed epilepsy controls. A standardized neuropsychological battery was performed at baseline and at 6 months on NCC cases and controls. A brain MRI was performed in patients with NCC at baseline and 6 months. Neuropsychological results were compared between NCC cases and controls at both time points. At baseline, patients with NCC had lower scores on attention tasks ( $p < 0.04$ ) compared with epilepsy controls but no significant differences compared to healthy controls. Six months after receiving anti-parasitic treatment, the NCC group significantly improved on tasks involving psychomotor speed ( $p < 0.02$ ). QoL at baseline suggested impaired mental function and social function in both the NCC and epilepsy group compared with healthy controls. QoL gains in social function ( $p = 0.006$ ) were noted at 6 months in patients with NCC.

**Conclusions/Significance** Newly diagnosed patients with NCC in this sample had mild cognitive deficits and more marked decreases in quality of life at baseline compared with controls. Improvements were found in both cognitive status and quality of life in patients with NCC after treatment.

**Author Summary** Neurocysticercosis (NCC) is one of the most common parasitic infections of the central nervous system. Cognitive changes have been frequently reported with this disease but have not been well studied. Our study team recruited a group of new onset NCC cases and a matched set of healthy neighborhood controls and new onset epilepsy controls in Lima, Peru for this study. A neuropsychological battery was administered at baseline and at 6 months to all groups. Brain MRI studies were also obtained on NCC cases at baseline and at 6 months. Newly diagnosed patients with NCC had mild cognitive deficits and more marked decreases in quality of life at baseline compared with controls. Improvements were found in both cognitive status and quality of life in patients with NCC after treatment. This study is the first to assess cognitive status and quality of life longitudinally in patients with NCC and provides new data on an important clinical morbidity outcome.